

ASHVIN SINGH

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SUMMARY

B.Tech Computer Science (AIML) student with hands-on experience in software development, machine learning, and deep learning. Has built and shipped production-grade AI tools and full-stack web applications integrating LLMs, RESTful APIs, and cloud services. Proficient in Python, TensorFlow, PyTorch, and modern backend frameworks; seeking to leverage these skills in applied AI and software engineering.

EDUCATION

VIT Bhopal University

B.Tech in Computer Science & Engineering (AIML) – GPA: 8.18/10.0

- Relevant Coursework: Data Science, Machine Learning, Artificial Intelligence, Deep Learning, DBMS, Data Structures & Algorithms

Prayagraj, UP, India

Expected Aug. 2027

TECHNICAL SKILLS

Languages: Python, C++, C, JavaScript

AI / ML / Deep Learning: TensorFlow, PyTorch, Scikit-Learn, Pandas, NumPy, CNNs, Computer Vision, NLP, Collaborative Filtering, SVD

Backend & APIs: FastAPI, SQLAlchemy, SQLite, RESTful API Design, OpenRouter API, Gmail API, Google OAuth 2.0

Frontend & Tools: React, Vite, HTML, CSS, Git, GitHub, Linux, Streamlit, Jupyter Notebook

Concepts: Object-Oriented Programming, DBMS, Version Control, LLM Integration, Agile Development

PROJECTS

Inbotic | *FastAPI, SQLAlchemy, React, Gmail API, Google Tasks API, LLM, SQLite*

Jan. 2026

- Architected a full-stack multi-user web app that converts Gmail messages into actionable Google Tasks using LLM-assisted NLP, validated with up to 10 users.
- Engineered a FastAPI backend spanning 8 Python modules with Google OAuth 2.0 authentication, rate limiting, and 2 interface layers (Jinja2 and React/Vite SPA).
- Connected 3 Google APIs (Gmail, Tasks, OAuth 2.0) end-to-end with SQLite-backed persistence for user, task, and email state.

AI Terminal | *Python, OpenRouter API, LLM Integration, Shell Scripting, python-dotenv*

Mar. 2025

- Developed a cross-platform CLI AI assistant (Windows/macOS/Linux) that converts natural language to shell commands via LLM API, with automatic OS and shell detection.
- Implemented a 3-model fallback chain for high availability, a 50-entry JSON conversation history, and pre-execution safety prompts blocking destructive commands.
- Shipped `setup.sh` and `setup.bat` scripts reducing installation time to under 2 minutes on all 3 platforms.

Movie Recommendation System | *Python, Surprise (SVD/KNN), Pandas, NumPy, Streamlit*

Apr. 2025

- Constructed a collaborative filtering engine using SVD matrix factorization that achieves 90% prediction accuracy on 1,200+ user ratings, with an interactive Streamlit interface.

BaseMem | *Python, SQLite, MCP, Flask, D3.js, tree-sitter*

Apr. 2026

- Built a multi-interface AI knowledge base (CLI, MCP server, Flask REST API) that persists context across sessions for 5+ agent platforms including Claude Code, Codex, Cursor, Windsurf, and Gemini.
- Designed a weighted graph storage layer using SQLite + FTS5 with Jaccard similarity auto-linking, edge reinforcement, decay/pruning, and 3-tier memory management (hot/warm/compacted).
- Implemented 30 MCP tools covering graph traversal, full-text search, agent-driven reranking, and tree-sitter code intelligence indexing 300+ languages.

PROFESSIONAL EXPERIENCE

Core Member

Bhopal, India

College Tech Club, VIT Bhopal

Mar. 2024 – Mar. 2025

- Coordinated a university-wide hackathon across 3 judging rounds with 50+ participants; crafted Figma prototypes and improved average submission responsiveness by 30%.
- Led outreach campaigns across 4+ campus platforms, reaching 1,000+ students and increasing event participation.

EXTRACURRICULAR ACTIVITIES

Open Source Contributor

GitHub

Python AI & Web Framework Projects

2023 – Present

- Submitted 5+ contributions (bug fixes, documentation, and features) to Python-based open-source projects; earned the *Pull Shark* GitHub Achievement.
- Actively participated in hackathons, coding competitions, and data science challenges on Kaggle, LeetCode, and Codeforces.

CERTIFICATIONS

Machine Learning Specialization – Andrew Ng, Coursera

C++ Programming Certificate – Udemy